

BEST 'A' LEVEL BIOLOGY REVISION NOTES®

ATP structure and synthesis

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TOPIC 7: ENERGETICS

ATP

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (ATTITUDES, SKILLS AND KNOWLEDGE)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.6.1 ATP Structure and Synthesis	- outline the need for energy in living organisms - describe ATP structure as a phosphorylated nucleotide - describe synthesis of ATP by chemiosmosis	- Anabolic reactions - Active transport - Movement - Maintenance of body temperature - Structure of ATP - Chemiosmosis	- Discussing uses of energy. - Illustrating the structure of ATP. - Illustrating the chemiosmosis coupling of ATP synthesis.	- Print media - ICT tools - Braille software/Jaws - Model

1. Energy

The Need for Energy

- Energy is the ability/capacity to do work.
- Living organisms are composed of cells, and within each cell, **work (activities and processes)** is constantly being carried out to maintain life.
- Work in a living organism requires **energy** and usable carbon compounds.

TABLE: Essential work within organisms

Type of work	Examples
Transporting substances across membranes	- Active transport using the sodium–potassium pump in cell membranes - Exocytosis of digested bacteria from white blood cells
Anabolic reactions	- Synthesis of DNA from nucleotides - Synthesis of protein from amino acids
Movement	- Cellular movement of chromosomes via the spindle - Mechanical contraction of muscles
Maintaining body temperature	Only occurs in mammals and birds

- ❖ Energy is needed for all the activities of living organisms, such as **nutrition, excretion, sensitivity, movement and reproduction**. It is also required to **build and maintain body structures**, and to **maintain body temperature**.
- ❖ **Specifically, all organisms require energy for:**
 - **metabolism** especially **anabolism** (building up of complex molecules, including macromolecules) e.g. synthesis of proteins, lipids etc
 - **active transport of molecules and ions across membranes**, carried out by membrane pumps.
 - **Phagocytosis, pinocytosis, exocytosis, endocytosis.**

ATP

- **movement by organisms**, including those due to **muscle contractions**. Energy is also transferred for movements within cells and by cell organelles.
- **formation and secretion of substances from cells**, including enzymes and hormones.
- **maintenance, repair and division of cells and organelles**
- **electrical transmission of nerve impulses**.
- **maintenance of body temperature** by heat from respiration.
- **bioluminescence** – emission of light without heat by fireflies, glow worms and some marine organisms.

The source of energy & materials

- For nearly all organisms **the sun** is the primary source of energy
- The reactions of **photosynthesis** store energy in organic molecules
 - Light energy from the sun is transformed into chemical potential energy in the synthesis of carbohydrates
 - The carbohydrates formed are then used in the synthesis of ATP (from their breakdown) or are combined and modified to form all the usable organic molecules that are essential for all metabolic processes within the plant
 - Photosynthesis is carried out by the first organism in a food chain, such as plants and some other small organisms
- **Respiration** in all living cells releases energy from the breakdown of organic molecules
- Respiration involves the transfer of **chemical potential energy** from nutrient molecules (such as carbohydrates, fats and proteins) into a usable energy form (through the synthesis of **ATP**) that can be used for work within an organism

Glucose equations

Glucose + oxygen → carbon dioxide + water + energy



- Autotrophs are organisms that are able to **synthesise their own usable carbon compounds** from carbon dioxide in the atmosphere through photosynthesis
- Heterotrophs don't have this ability. They require a supply of pre-made usable carbon compounds which they get from their food

Exam Tip

- According to the laws of thermodynamics, energy cannot be created or destroyed; it is **transformed** from one form into another. Be careful not to say that energy is “created” when talking about photosynthesis and respiration.
- You may also be expected to name examples of energy-requiring reactions in organisms:
- The **sodium-potassium pump** that is found on many cell membranes is a great example of active transport. Three sodium ions are taken out of the cell while two potassium ions are taken in, both against their respective concentration gradients
- The movement and contraction of **muscles** also requires substantial amounts of energy

2. ATP

ATP: Universal Energy Currency

- Energy released during the reactions of respiration is transferred to the molecule **adenosine triphosphate** (ATP)
- ATP is a small and soluble molecule that provides a **short-term store** of chemical energy that cells can use to do work
- It is vital in linking energy-requiring and energy-yielding reactions
- ATP is described as a universal energy currency
 - Universal: It is used in all organisms
 - Currency: Like money, it can be used for different purposes (reactions) and is reused countless times
- The use of ATP as an 'energy-currency' is beneficial for many reasons:
 - The hydrolysis of ATP can be carried out quickly and easily wherever energy is required within the cell by the action of just one enzyme, ATPase
 - A useful (not too small, not too large) quantity of energy is released from the hydrolysis of one ATP molecule – this is beneficial as it reduces waste but also gives the cell control over what processes occur
 - ATP is relatively stable at cellular pH levels

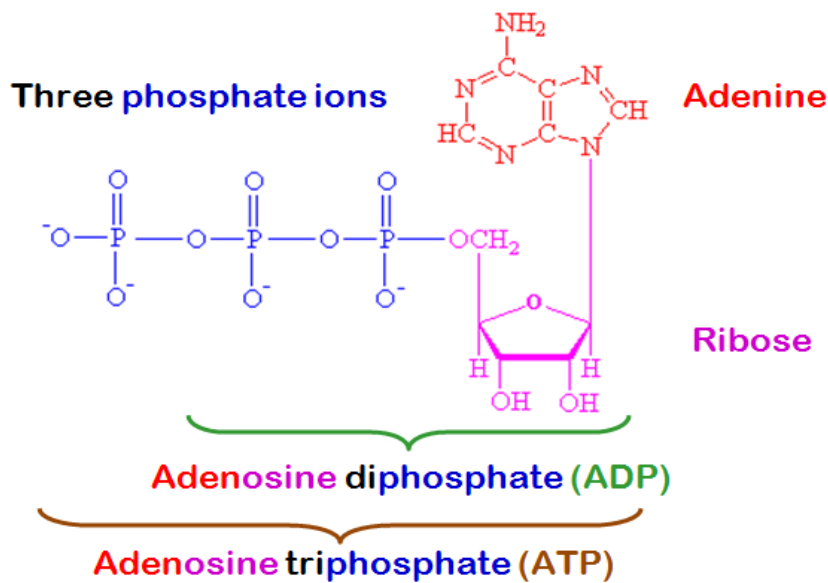
Question: Outline why ATP is an efficient energy currency molecule.

- Features of ATP that make it suitable as the universal energy currency:
 - Loss of phosphate / hydrolysis, leads to energy release
 - Small packets of energy
 - Small / water-soluble, so can move around cell
 - Immediate energy donor
 - Acts as link between energy-yielding and energy-requiring reactions
 - High turnover

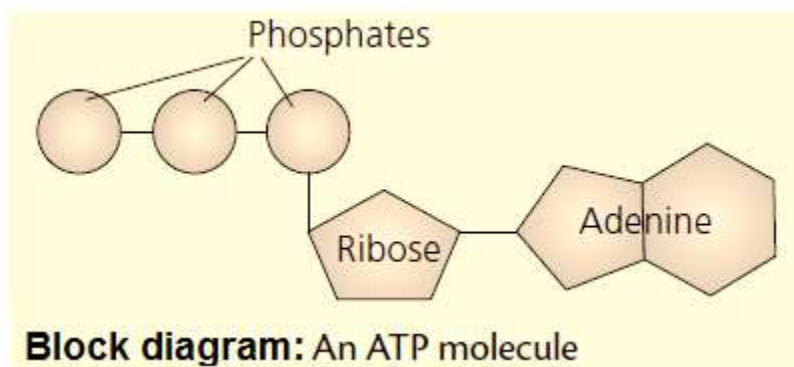
Structure of ATP

- ATP is a phosphorylated **nucleotide**
- It is made up of:
 - Ribose sugar
 - Adenine base
 - Three phosphate groups
- The following diagram shows the structure of an ATP molecule

ATP



- The diagram below shows the simplified block diagram of an ATP molecule.

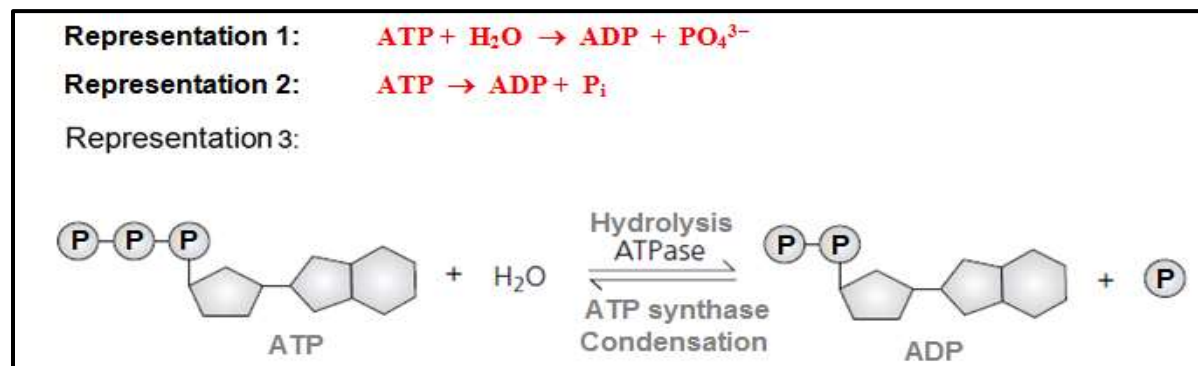


Hydrolysis of ATP

Catalysed by **the enzyme ATP hydrolase (ATPase)**, the **hydrolysis** of **ATP** produces:

- **adenosine diphosphate (ADP)**
- **a phosphate group (PO_4^{3-} often represented as P_i)**
- **and releases ENERGY**

This hydrolysis can be represented in a number of ways:



ATP

So when **ATP is hydrolysed**, **water is added** and the **phosphate group leaves**. Water hydrolyses the bond and the phosphate group leaves. This **gives out about 30.5 kJ per mol of energy**. So you have gone from **ATP** to **ADP** and **P_i** (inorganic phosphate).

- As ADP forms **free energy** is released that can be used for processes within a cell e.g. DNA synthesis
 - Removal of one phosphate group from ATP releases 30.8 kJ mol^{-1} of energy, forming ADP
 - Removal of a second phosphate group from ADP also releases 30.8 kJ mol^{-1} of energy, forming AMP
 - Removal of the third and final phosphate group from AMP releases 14.2 kJ mol^{-1} of energy, forming adenosine

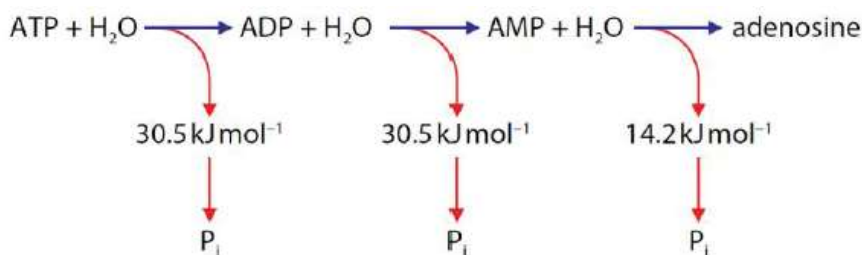
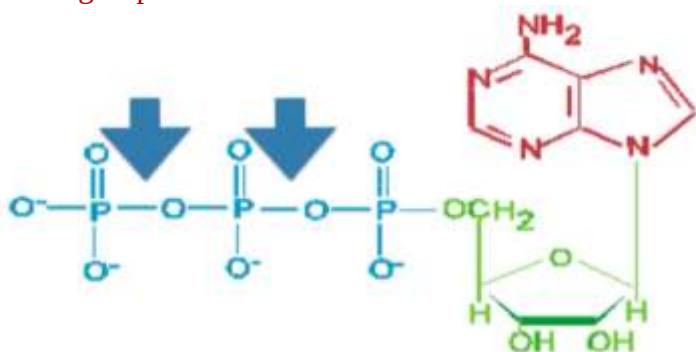


TABLE: Features of ATP.

Feature	Benefit
Releases a small but sufficient amount of energy (75.8 kJ mol^{-1} from the complete hydrolysis of ATP)	This is enough energy to drive important metabolic reactions while keeping energy wastage low
Exists as a stable molecule	It doesn't break down unless a catalyst (ATPase) is present so energy won't be wasted
Can be recycled	The breakdown of ATP is a reversible reaction, ATP can be reformed from ADP and P _i . This means the same molecule can be reused elsewhere in the cell for different reactions
Hydrolysis is quick and easy	Allows cells to respond to a sudden increase in energy demand
Soluble and moves easily within cells	Can transport energy to different areas of the cell
Forms phosphorylated intermediates	This can make metabolites more reactive and lower the activation energy required for a reaction

HOW DOES ATP STORE ENERGY?

- ❖ ATP stores energy as chemical potential energy in the bonds between its phosphate groups.



Exactly how does ATP store this energy?

ATP has 3 **phosphate groups**. Each **phosphate group is negatively charged so they repel one another**. **This makes the covalent bonds that link them rather unstable**. It's like a **coiled spring**. Now these **unstable covalent bonds** are **high energy bonds** and have **low activation energy**. What does this mean- **because of the low activation energy they are easily broken** and **release energy** in the process.

- ❖ **Instability of its Phosphate bonds makes ATP an excellent energy donor.**

Exam Tip

- Be careful not to use the terms energy and ATP interchangeably. Energy is the capacity to do work. ATP is a molecule which stores (chemical potential) energy and carries it to places in the cell that need energy to do work.
- For example, saying that “energy is used in muscles” in the exam won't get you marks for describing how the muscles work as it is too vague an answer.

ATP Synthesis

- On average humans use more than 50 kg of ATP in a day but only have a maximum of ~ 200g of ATP in their body at any given time
- Organisms **cannot build up large stores of ATP** and it rarely passes through the cell surface membrane
- This means the cells must make ATP as and when they need it
- ATP is formed when ADP is combined with an inorganic phosphate (Pi) group
 - This is an **energy-requiring reaction**
 - **Water is released** as a waste product (therefore ATP synthesis is a condensation reaction)

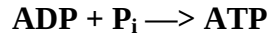
Types of ATP synthesis

- ATP is made during the reactions of respiration and photosynthesis
 - All of an animal's ATP comes from respiration
- ATP can be made in two different ways:
 - **Substrate-linked phosphorylation**
 - **Chemiosmosis**

Substrate-linked phosphorylation

ATP

- ATP is formed by **transferring a phosphate directly from a substrate molecule to ADP**



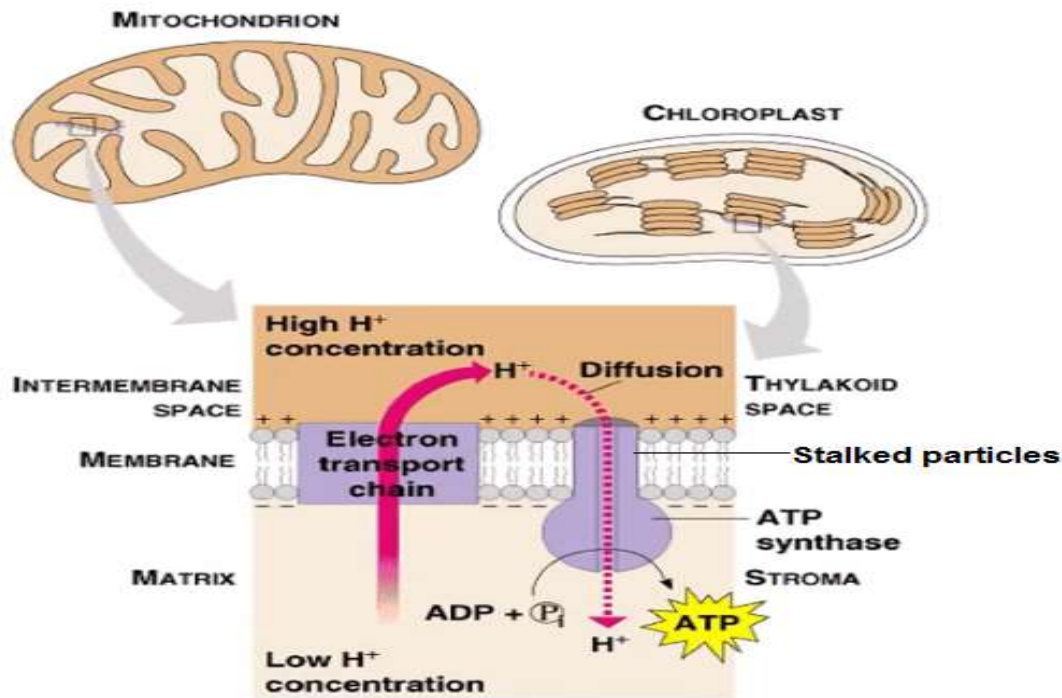
- The energy required for the reaction is **provided directly by another chemical reaction**
- This type of ATP synthesis occurs in the cell cytoplasm and in the matrix of the mitochondria
- It only accounts for a small amount of the ATP synthesised during aerobic respiration
 - ~ 4 / 6 ATP per glucose molecule
- This type of ATP synthesis takes place in glycolysis.

Chemiosmosis

- This specific type of ATP synthesis involves a proton (hydrogen ion) gradient across a membrane
- It takes place across the inner membrane of the mitochondria and the thylakoid membrane of chloroplasts
- An electron transport chain helps to establish the **proton concentration gradient**
 - High energy electrons move from carrier to carrier releasing energy that is used to pump protons (up a concentration gradient) across the inner membrane into the intermembrane space
 - Protons are pumped from a low concentration in the mitochondrial matrix to a high concentration in the intermembrane space
- The protons then move down the concentration gradient into the matrix which **releases energy**
- The protons move through the **ATP synthase complex** which uses the released energy to **drive the phosphorylation of ATP**
- Oxygen acts as the **final electron and proton acceptor** to form water
- Most of the ATP made during respiration is synthesised via chemiosmosis
 - ~ 32 / 34 ATP per glucose molecule

Question: Describe the synthesis of ATP by chemiosmosis.

So, **HOW IS ATP SYNTHESISED BY CHEMIOSMOSIS?**



Energy is released during the transport of electrons along the ETC (electron transport chain). There is going to be a difference in the H^+ concentration on either side of the membrane. And this occurs in both chloroplasts and mitochondria. If you look at the diagram you will see that the H^+ concentration is higher in the thylakoid space of the chloroplast or the intermembrane space of the mitochondrion than in the stroma or matrix respectively. So what's going to happen? The H^+ ions will flow down their concentration gradient but they can only go through a special protein channel and we call this channel a stalked protein particle. So when they flow through this channel, electrical potential energy is released. And this is used to synthesise ATP. We call this whole process the chemiosmotic theory of ATP synthesis. At the bulging base of the stalked particle there is an enzyme called ATP synthase (a.k.a. ATP synthetase). This catalyses the phosphorylation reaction so that ATP is formed.

We are going to discuss this in detail when we look at oxidative phosphorylation

TABLE: ATP synthesis

	Substrate-linked phosphorylation	Chemiosmosis
Process	The phosphate of a substrate molecule is directly transferred to ADP to form ATP It uses energy directly provided by another chemical reaction	The energy released by the movement of hydrogen ions down a concentration gradient is used to synthesise ATP via the enzyme ATP synthase Oxygen acts as the final hydrogen/electron acceptor .
Location	Cytoplasm of cells/ Matrix of mitochondria	Inner mitochondrial membrane/ Thylakoid membrane of chloroplasts
Quantity of ATP produced during respiration	Small (4/6 per glucose molecule)	Large (34/32 per glucose molecule)

Exam Tip

You may be asked to identify which type of ATP synthesis is occurring at different stages of respiration and photosynthesis. Remember that chemiosmosis involves a **proton gradient** that has been created by an **electron transport chain** and it takes place across an **inner membrane**.

Examination style questions

1. Describe the structure and synthesis of ATP and its universal role as the energy currency in all living organisms. [8]

- nucleotide ;
- adenine + ribose / pentose + three phosphates ;
- loss of phosphate leads to energy release / hydrolysis releases
- 30.5 kJ ;
- $\text{ADP} + \text{P}_i \leftrightarrow \text{ATP}$ (reversible reaction) ;
- synthesised during, glycolysis / Krebs cycle / substrate level phosphorylation ;
- synthesised, using electron carriers / oxidative phosphorylation /
- photophosphorylation ;
- in, mitochondria / chloroplasts ;
- ATP synthase / ATP synthetase ;
- chemiosmosis / description;
- used by cells as immediate energy donor ;
- link between energy yielding and energy requiring reactions / AW ;
- active transport / muscle contraction / Calvin cycle / protein synthesis

2. Fig. 1.1 shows the structure of ATP.

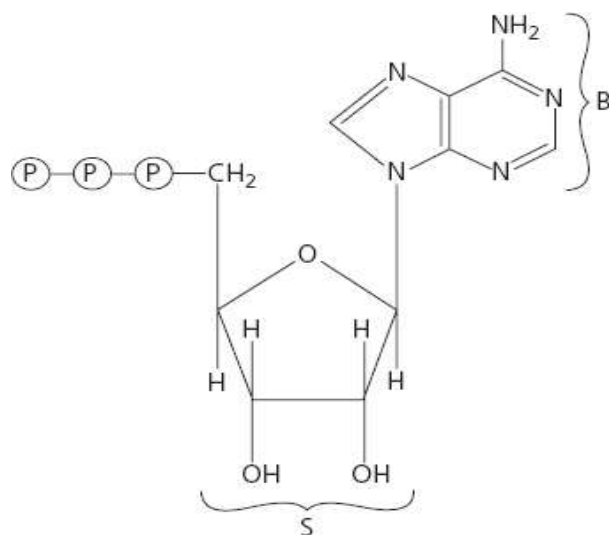


Fig. 1.1

a) i) Name the nitrogenous base labelled B.

[1]

Adenine

ii) Name the sugar labelled S.

[1]

Ribose;

Reject pentose

b) ATP is described as having a universal role as the energy currency in all living organisms. Explain why it is described in this way.

[4]

1 *energy is released when it is hydrolysed*; A equation A joules for energy

2 *easily hydrolysed*;

3 *(energy) used in, processes / reactions*; A named process

4 *rapid turnover*;

5 *links catabolic and anabolic reactions* / AW ;

6 *found in, most cells / all organisms*;

7 *soluble so easily moved (within cell)*;

8 *ATP produced from variety of reactions*; A named reactions

c) State precisely two places where ATP is synthesised in cells.

[2]

1 *ETC (electron transport chain)/ inner mitochondrial membrane / crista / stalked particles*;

2 *grana / thylakoids / inner chloroplast membrane*;

3 *cytoplasm / cytosol*;

4 *mitochondrial matrix*;

[Total: 8]

(Cambridge International AS and A Level Biology 9700, Paper 04 Q6 Nov 2008)